Migraine prophylaxis 2024 evidence?

Evidence searched on 2026-10-09 (PubMed, ClinicalTrials.gov, FDA drug labels, web)

Clinical summary

Bottom line Migraine prophylaxis in 2024 is supported by established oral drugs (e.g., beta-blockers, amitriptyline, topiramate) that reduce monthly migraine days by ~1–1.4 days vs placebo, as well as emerging CGRP-targeted therapies (gepants) and procedural options like ultrasound-guided greater occipital nerve block (GONB), which significantly lowers pain intensity and headache frequency. However, most emerging pharmacological targets remain preclinical, and phenotype-stratified approaches (e.g., for vestibular migraine) require further validation.

Most important recent finding Ultrasound-guided greater occipital nerve block (US-GONB) reduced monthly headache frequency by **9.12 days** (95% CI –11.65 to –6.58) in a meta-analysis of 658 patients [1].

Immediate context Established oral prophylactic agents (e.g., beta-blockers, tricyclics, topiramate) are guideline-supported for episodic migraine, while CGRP antagonists (gepants) and monoclonal antibodies are newer but cost/reimbursement barriers persist. Vestibular migraine management lacks robust phenotype-stratified evidence, and most non-pharmacological interventions remain exploratory.

Top treatment drugs (established first, then the most notable latest drug)

Drug	Indication named in the FDA label	Source
Valproic Acid	"For prophylaxis of migraine headaches, valproate is contraindicated in women who are pregnant and in women of childbearing potential who are not using effective contraception."	DailyMed

Latest findings

- **High-dose omega-3 fatty acids (EPA/DHA)** (2024, Network Meta-Analysis)

In a network meta-analysis of 11 interventions for migraine prophylaxis, high-dose EPA/DHA supplementation demonstrated the largest reduction in migraine days per month (−2.32, 95% CI −3.32 to −1.32) and migraine severity (SMD −2.23, 95% CI −3.17 to −1.30) compared with placebo. The study highlights its favorable acceptability profile, though confidence intervals for acceptability were wide (odds ratio 1.00, 95% CI 0.06 to 17.41) [2].

- **Melatonin** (2026, Meta-Analysis)

A systematic review of 9 randomized controlled trials (n = 788) found melatonin reduced migraine attack duration by 4.98 hours (MD −4.98, 95% CI −9.30 to −0.67) and headache days by 1.54 days (MD −1.54, 95% CI −2.50 to −0.58) versus placebo.

- **Rimegepant for preventive treatment** (2025, Randomized Controlled Trial)

Adverse events were mild, with no liver toxicity signals [3].

- **Atogepant for refractory episodic migraine** (2024, Randomized Controlled Trial)

In the ELEVATE trial (n = 313), atogepant reduced monthly migraine days by 2.4 days (95% CI −3.2 to −1.5) versus placebo in patients who failed conventional oral prophylaxis. Constipation was the most common adverse event (10% vs. 3% with placebo) [4].

- **Ultrasound-guided greater occipital nerve block (US-GONB)** (2026, Meta-Analysis, advance publication)

A systematic review of 12 studies (n = 658) found US-GONB reduced monthly headache frequency by 9.12 days (MD −9.12, 95% CI −11.65 to −6.58) and analgesic use by 10.01 units (MD −10.01, 95% CI −12.91 to −7.11). Proximal injections showed numerically larger reductions than distal approaches, though adverse events (e.g., sensory loss) were transient [1].

Current evidence

Approach	Evidence level	What the evidence shows	Reference
CGRP monoclonal antibodies (e.g., galcanezumab, erenumab, fremanezumab, eptinezumab)	Established	Reduced monthly migraine days by 1.1–3.8 (vs. placebo) in episodic and chronic migraine; high-confidence evidence supports efficacy.	[5]; [6]
Atogepant (oral CGRP receptor antagonist)	Established	Reduced monthly migraine days by 0.7–2.4 (vs. placebo) in episodic and chronic migraine; moderate-to-high confidence in efficacy.	[5]; [4]
Rimegepant (acute/short-term preventive)	Emerging	Reduced monthly migraine days by 1.1 (vs. placebo) in a Phase III trial; well-tolerated with mild adverse events.	[3]
Melatonin	Emerging	Reduced migraine attack duration by 4.98 hours and headache days by 1.54 (vs. placebo) in meta-analysis; favorable safety profile; adjunctive role during titration of first-line agents.	[7]
Ultrasound-guided greater occipital nerve block (US-GONB)	Experimental or preliminary	Reduced pain intensity (MD -3.48), monthly headache frequency (MD -9.12 days), and analgesic use; proximal injections showed larger reductions; mild transient adverse events.	[1]
Topiramate	Established	Reduced migraine days by 1.2–1.4 (vs. placebo) in episodic migraine; contraindicated in pregnancy; caution in women of childbearing age due to neurodevelopment risks.	[8]; [9]
High-dose omega-3 fatty acids (EPA/DHA)	Emerging	Largest reduction in migraine frequency (ratio of means 0.38 vs. placebo) and severity; highest acceptability among studied prophylactic treatments.	[2]

Treatment Drug Landscape

Drugs used for treatment of this condition

Approved: an FDA label names this condition. Guideline-recommended: a guideline organisation's page recommends the drug. Doses appear only where an FDA label or source gives them for the use shown; individual dosing is a prescriber's decision.

Drug	Indication	Dose / route (if sourced)	Approval status / date	Key evidence
Valproic Acid	"For prophylaxis of migraine headaches, valproate is contraindicated in women who are pregnant and in women of childbearing potential who are not using effective contraception."	Route: Oral. No dosing section for this condition was found in the retrieved label text; see the linked label.	FDA-approved for the use shown; label effective 2024-05-27; original approval date not retrieved	FDA label. DailyMed label

Other drugs: used for symptoms (not established treatments of this condition)

Drugs that a retrieved review, guideline or web page describes as used for the condition, its symptoms or co-occurring conditions, for which no FDA label for this use was retrieved. Drugs still being tested in trials are listed in the Clinical trials section below.

Drug	Indication	Key evidence
topiramate	chronic migraine headache/migraine days (population: adults with chronic migraine)	"RESULTS: We found 51 individual articles, reporting 11 randomised controlled trials, testing 6 drugs (topiramate, Botox, eptinezumab, erenumab, fremanezumab, galcanezumab), versus placebo, on 7352 adults with chronic migraine." [6] (Mistry et al., Health technology assessment (Winchester, England), 2024)
Botox (onabotulinumtoxina)	chronic migraine headache/migraine days (population: adults with chronic migraine)	"RESULTS: We found 51 individual articles, reporting 11 randomised controlled trials, testing 6 drugs (topiramate, Botox, eptinezumab, erenumab, fremanezumab, galcanezumab), versus placebo, on 7352 adults with chronic migraine." [6] (Mistry et al., Health technology assessment (Winchester, England), 2024)
atogepant	as stated in the quoted source sentence	"High-level evidence also supports the efficacy of the CGRP receptor antagonist atogepant (0.7-2.4 fewer migraine days per month) in both episodic and chronic migraine." [8]
eptinezumab	as stated in the quoted source sentence	"Monoclonal antibodies against calcitonin gene-related peptide (CGRP) or the CGRP receptor are effective against both episodic and chronic migraine: Eptinezumab lowered the number of migraine days per month by 0.7-3.2, fremanezumab by 1.3-3.8, galcanezumab by 1.1-3.7, and erenumab by 1.0-2.5." [8]
fremanezumab	as stated in the quoted source sentence	"Monoclonal antibodies against calcitonin gene-related peptide (CGRP) or the CGRP receptor are effective against both episodic and chronic migraine: Eptinezumab lowered the number of migraine days per month by 0.7-3.2, fremanezumab by 1.3-3.8, galcanezumab by 1.1-3.7, and erenumab by 1.0-2.5." [8]
galcanezumab	as stated in the quoted source sentence	"Monoclonal antibodies against calcitonin gene-related peptide (CGRP) or the CGRP receptor are effective against both episodic and chronic migraine: Eptinezumab lowered the number of migraine days per month by 0.7-3.2, fremanezumab by 1.3-3.8, galcanezumab by 1.1-3.7, and erenumab by 1.0-2.5." [8]
erenumab	as stated in the quoted source sentence	"Monoclonal antibodies against calcitonin gene-related peptide (CGRP) or the CGRP receptor are effective against both episodic and chronic migraine: Eptinezumab lowered the number of migraine days per month by 0.7-3.2, fremanezumab by 1.3-3.8, galcanezumab by 1.1-3.7, and erenumab by 1.0-2.5." [8]
rimegepant	as stated in the quoted source sentence (population: adults)	"RESULTS: Of 425 adults, 152 used rimegepant for STP (female, n = 98 [64.5%]; male, n = 54 [35.5%])." [10]

Recently approved or updated (FDA, EMA and CDSCO announcements)

No FDA, EMA or CDSCO announcement dated within the last 3 years was found for this question. This does not mean none exists.

2 further regulator page(s) were read but are not listed: the page states no date, or a date more than 3 years old, so they cannot be shown as recent.

Safety warnings (FDA boxed warnings)

Drug(s) (FDA label)	Boxed warning, in the label's words
Botox (onabotulinumtoxina)	"DISTANT SPREAD OF TOXIN EFFECT Postmarketing reports indicate that the effects of BOTOX and all botulinum toxin products may spread from the area of injection to produce symptoms consistent with botulinum toxin effects. These may include asthenia, generalized muscle weakness, diplopia, ptosis, dysphagia, dysphonia, dysarthria, urinary incontinence and breathing difficulties. These symptoms have been reported hours to weeks after injection."
Valproic Acid	"Hepatotoxicity General Population: Hepatic failure resulting in fatalities has occurred in patients receiving valproate and its derivatives. These incidents usually have occurred during the first six months of treatment. Serious or fatal hepatotoxicity may be preceded by non-specific symptoms such as malaise, weakness, lethargy, facial edema, anorexia, and vomiting. In patients with epilepsy, a loss of seizure control may also occur."

These are the labels' own boxed warnings, not a complete safety review: read the full label. No boxed warning on the retrieved label of: topiramate, atogepant, eptinezumab, fremanezumab, galcanezumab, erenumab, rimegepant.

Clinical trials

Trial	Intervention	Phase	Status (last update)	What the record shows
NCT06810505	Atogepant; Placebo for Atogepant	Phase 3	Recruiting (2026-10-09)	Under investigation; results not yet available; effectiveness cannot be inferred.
NCT07568496	Lidocaine 2% and Methylprednisolone 80 mg; Placebo	Phase not applicable	Recruiting (2026-10-08)	Under investigation; results not yet available; effectiveness cannot be inferred.
NCT07629635	Prasugrel 5mg; Transcatheter Closure of Patent Foramen Ovale	Phase 3	Recruiting (2026-10-08)	Under investigation; results not yet available; effectiveness cannot be inferred.
NCT07849283	•Group A: Patients will receive Medical treatment only: Beta-blockers (Propranolol) , ...	Phase not applicable	Not yet recruiting (2026-10-07)	Under investigation; results not yet available; effectiveness cannot be inferred.
NCT07816367	Elismetrep; Placebo	Phase 3	Recruiting (2026-10-07)	Under investigation; results not yet available; effectiveness cannot be inferred.
NCT07861412	Patient Headache Diary; Patient Reported Outcome	Phase not applicable	Not yet recruiting (2026-10-06)	Under investigation; results not yet available; effectiveness cannot be inferred.
NCT05711394	Atogepant; Placebo-Matching Atogepant	Phase 3	Recruiting (2026-10-06)	Under investigation; results not yet available; effectiveness cannot be inferred.
NCT06792500	Glycerol Tributyrate	Phase 1/2	Not yet recruiting (2026-10-05)	Under investigation; results not yet available; effectiveness cannot be inferred.
NCT05983354	Oral metoclopramide and ibuprofen; Intravenous metoclopramide and ketorolac	Phase not applicable	Withdrawn (2026-10-05)	Registry status WITHDRAWN; no effectiveness conclusion.

Trial	Intervention	Phase	Status (last update)	What the record shows
NCT07616128	EVO756, Dose 1; EVO756, Dose 2	Phase 2	Recruiting (2026-10-02)	Under investigation; results not yet available; effectiveness cannot be inferred.
NCT07856056	Nortriptyline; Amitriptyline	Phase 4	Not yet recruiting (2026-10-02)	Under investigation; results not yet available; effectiveness cannot be inferred.
NCT07361549	Lidocaine (drug); Saline	Phase 2	Recruiting (2026-10-01)	Under investigation; results not yet available; effectiveness cannot be inferred.

Registry status shows that research is being conducted, not that a treatment works. Registry status is separate from published results.

Key studies

1. **Pringsheim et al. (2026)**, Systematic Review: For episodic migraine, high-confidence evidence showed that galcanezumab and erenumab are more effective than placebo in reducing headache frequency. Moderate-confidence evidence supported benefit from atogepant, eptinezumab, fremanezumab, propranolol, topiramate, and valproate [5].
2. **Tassorelli et al. (2024)**, Randomized Controlled Trial: In adults with episodic migraine, atogepant reduced monthly migraine days by 4.2 (vs. 1.9 with placebo) over 12 weeks, with constipation as the most common adverse event [4].
3. **Mistry et al. (2024)**, Systematic Review/Network Meta-Analysis: In adults with chronic migraine, calcitonin gene-related peptide monoclonal antibodies (e.g., eptinezumab, fremanezumab) reduced headache/migraine days by 2.0–2.5 per month, outperforming topiramate and onabotulinumtoxinA [6].
4. **Diener et al. (2026)**, Review: Oral prophylactic drugs (e.g., beta-blockers, flunarizine, amitriptyline, topiramate) reduced migraine days/month by 0.44–1.4 vs. placebo. CGRP monoclonal antibodies (e.g., eptinezumab, fremanezumab) reduced days by 0.7–3.8 [8].
5. **Abouelmagd et al. (2026)**, Meta-Analysis: Melatonin reduced migraine attack duration by 4.98 hours and headache days by 1.54 vs. placebo in randomized trials (n=788), with a favorable safety profile [7].

Conflicting evidence

No direct disagreement was retrieved between studies on migraine prophylaxis efficacy for **adults**. However, **topiramate's role in women of childbearing age** is contested: while it remains supported by moderate-confidence evidence for chronic migraine [5], recent pharmacovigilance data [9] contraindicate its use during pregnancy and caution its use in non-pregnant women due to neurodevelopmental risks. No other direct contradictions in efficacy or safety were identified.

What remains under investigation

- **Efficacy of high-dose EPA/DHA** in pediatric migraine prophylaxis: A 2024 network meta-analysis (Tseng et al.) reported compelling efficacy in adults, but **no direct pediatric evidence** was retrieved.
- **Long-term safety and cost-effectiveness** of gepants (e.g., atogepant, rimegepant) in real-world settings: While short-term efficacy is supported by limited evidence [11, 3], **long-term data (beyond 12 weeks)** and comparative cost-effectiveness remain understudied.
- **Head-to-head trials** comparing melatonin vs. standard prophylactic agents (e.g., amitriptyline) for migraine frequency reduction: Abouelmagd et al. (2026) highlighted melatonin's benefits but noted **no direct comparisons** with first-line agents.

- **Ultrasound-guided greater occipital nerve block (US-GONB) mechanisms**: Sabet et al. (2026) demonstrated efficacy in reducing headache frequency, but **no mechanistic trials** explain why proximal vs. distal injections differ in effect.
- **Feasibility of short-term prevention (STP) strategies** (e.g., rimegepant for perimenstrual migraine): Monteith et al. (2026) described provider perceptions but **no randomized trials** validate STP efficacy vs. traditional prophylaxis.
- **Efficacy of orexinergic pathways or nitric oxide synthase inhibitors** in migraine: Atraszkiewicz et al. (2026) noted mechanistic plausibility but **no clinical trials** demonstrating efficacy were retrieved.

Evidence limitations

- Findings are drawn from abstracts, registry records and web pages, not full texts; study quality was not appraised from the full papers.
- Doses appear only where an FDA label gives them for this condition (section 2.1); for any other use, see a prescribing source. Approval dates appear only where a retrieved source states them.
- 4 retrieved record(s) were excluded because the relevance reading judged them not to answer the question.

Sources

1. Sabet H, Samir A, Lashin B, et al. Exploring Proximal and Distal Ultrasound-Guided Greater Occipital Nerve Block Approaches in Migraine: A Systematic Review and Meta-Analysis. *Pain practice : the official journal of World Institute of Pain*. 2026-11-01. PMID 42743099. DOI 10.1111/papr.70202.
2. Tseng PT, Zeng BY, Chen JJ, et al. High Dosage Omega-3 Fatty Acids Outperform Existing Pharmacological Options for Migraine Prophylaxis: A Network Meta-Analysis. *Advances in nutrition (Bethesda, Md.)*. 2024 Feb. PMID 38110000. DOI 10.1016/j.advnut.2023.100163.
3. Kitamura S, Matsumori Y, Yamamoto T, et al. Efficacy and safety of rimegepant for the preventive treatment of migraine in Japan: A double-blind, randomized controlled trial. *Headache*. 2025 Sep. PMID 40542538. DOI 10.1111/head.14995.
4. Tassorelli C, Nagy K, Pozo-Rosich P, et al. Safety and efficacy of atogepant for the preventive treatment of episodic migraine in adults for whom conventional oral preventive treatments have failed (ELEVATE): a randomised, placebo-controlled, phase 3b trial. *The Lancet. Neurology*. 2024 Apr. PMID 38364831. DOI 10.1016/S1474-4422(24)00025-5.
5. Pringsheim T, Smith DB, Tanveer S, et al. Systematic Review of Pharmacologic Treatment for Migraine Prevention in Adults: Report of the AAN Guidelines Subcommittee and the American Headache Society. *Neurology*. 2026 Oct 13. PMID 42673559. DOI 10.1212/WNL.0000000000218112.
6. Mistry H, Naghdi S, Brown A, et al. Preventive drug treatments for adults with chronic migraine: a systematic review with economic modelling. *Health technology assessment (Winchester, England)*. 2024 Oct. PMID 39365169. DOI 10.3310/AYWA5297.
7. Abouelmagd ME, Aldemerdash MA, Khatatbeh AA, et al. Efficacy and Safety of Melatonin in Migraine Prophylaxis: A Systematic Review and Meta-Analysis of Randomized Controlled Trials. *Current pain and headache reports*. 2026-02-02. PMID 41627537. DOI 10.1007/s11916-025-01461-5.
8. Diener HC, Grans J, Reuter U, et al. Drugs for Migraine Prophylaxis. *Deutsches Arzteblatt international*. 2026-03-06. PMID 41572863. DOI 10.3238/arztebl.m2025.0234.
9. Moisset X, Demarquay G, de Gaalon S, et al. Migraine treatment: Position paper of the French Headache Society. *Revue neurologique*. 2024 Dec. PMID 39406556. DOI 10.1016/j.neurol.2024.09.008.
10. Monteith T, Cirillo J, Nakajima K, et al. Sex differences in the real-world use of rimegepant for short-term prevention of migraine in the United States. *Frontiers in neurology*. 2026. PMID 42851312. DOI

10.3389/fneur.2026.1901123.

11. Gautier A, Gicquel T, Mercerolle M, et al. Efficacy and safety of gepants in migraine management: a narrative review focusing on atogepant and rimegepant. *Journal of oral & facial pain and headache*. 2026-07-12. PMID 42548099. DOI 10.22514/jofph.2026.046.

Other sources (regulators, guideline organisations, web pages, FDA labels)

- FDA drug label: Valproic Acid (label effective 2024-05-27)

Research support, not a substitute for clinical judgement. Some items could not be fully verified; the detail is in the saved verification record.